

CFD Benchmark Testing of Selected Laminar Flow Problems

M. Baris DOGRUOZ

October 7, 2002

A benchmark testing and numerical validation of the finite-volume code, *Fluent* will be presented. Benchmark problems are very useful in terms of testing, evaluating and judging CFD codes. Thus, in this study, five laminar flow problems are investigated to test the performance of *Fluent* from different aspects:

1. Developing Flow between Parallel Plates
2. Thermally Developing Slug Flow between Parallel Plates
3. Laminar Flow and Heat Transfer over a Backward-Facing Step
4. Mixed Convection Flow and Heat Transfer in a Vertical Backward Facing Step
5. Buoyancy-Driven Flow in a Square Enclosure

First two problems provide an opportunity to investigate entry flows, the third problem investigates the separation and reattachment regions, and last two problems investigate the Buoyancy effects. In addition, except for the first case, associated heat transfer problem with the fluid flow was also solved.

The selected problems demonstrate standard benchmark cases, which were solved by utilizing various finite-difference and finite-element techniques. For a given accuracy, a grid-independent study is performed for each problem and the final results are compared with those from the open literature.

References

- [1] Gartling, D., K., "A Test Problem for Outflow Boundary Conditions - Flow over a Backward-Facing Step," *International Journal for Numerical Methods in Fluids*, Vol. 11, pp. 953-967, 1990.
- [2] Choudhury, D., "A Numerical Study of Laminar Flow and Heat Transfer in a Backward-Facing Step using Fluent: A Finite-Volume CFD," *Proceedings of ASME, Heat Transfer Division-HTD, Benchmark Problems for Heat Transfer Codes*, Vol. 222, pp. 53-57, 1992.
- [3] McDonald, J., W., Denny, V., E., Mills, A., F., "Numerical Solutions for Navier-Stokes Equations in Inlet Regions," *Journal of Applied Mechanics*, Vol. 39 Ser E, n4, pp. 873-878, 1972.
- [4] Burmeister, L., C., "Convective Heat Transfer," *Wiley&Sons*, New York, NY, 1983.
- [5] Dyne, B., R., Pepper, D., W., Brueckner, F., P., "Mixed Convection in a Vertical Channel with a Backward-Facing Step: A Benchmark Problem," *Proceedings of ASME, Heat Transfer Division-HTD, Computational Aspects of Heat Transfer Benchmark Problems*, Vol. 258, pp. 49-56, 1993.
- [6] AbdulNour, B., S., Potter, M., C., "A Stable, Iterative Finite-Difference Procedure for The Navier-Stokes Equations," *Proceedings of ASME, Fluids Engineering Division-FED, Benchmark Test Cases for Computational Fluid Dynamics*, Vol. 93, pp. 15-19, 1990.
- [7] Chopin, T., R., "Mixed Convection Flow and Heat Transfer in a Vertical Backward Facing Step Using the Flotran CFD Program," *Proceedings of ASME, Heat Transfer Division-HTD, Computational Aspects of Heat Transfer Benchmark Problems*, Vol. 258, pp. 21-28, 1993.
- [8] Dyne, B., R., Heinrich, J., C., "Flow over a Backward-Facing Step: A Benchmark Problem for Laminar Flow with Heat Transfer," *Proceedings of ASME, Heat Transfer Division-HTD, Benchmark Problems for Heat Transfer Codes*, Vol. 222, pp. 73-76, 1992.
- [9] Patankar, S., V., "CFD Benchmark Problems for Electronic Packaging: Some Desirable Characteristics," *Proceedings of ASME, Heat Transfer Division-HTD, Solutions to CFD Benchmark Problems in Electronic Packaging*, Vol. 255, pp. 31-35, 1993.
- [10] Choudhury, D., Woolfe, A., E., "Computation of Laminar Forced and Mixed Convection in a Heated Vertical Duct with a Step," *Proceedings of ASME, Heat Transfer Division-HTD, Computational Aspects of Heat Transfer Benchmark Problems*, Vol. 258, pp. 29-36, 1993.

- [11] Hajiloo, A., Engelman, M., S., "Flow over a Backward-Facing Step with Heat Transfer: A Benchmark Solution," *Proceedings of ASME, Heat Transfer Division-HTD, Benchmark Problems for Heat Transfer Codes*, Vol. 222, pp. 7-12, 1992.
- [12] Choudhury, D., "A FLUENT Simulation of Bouyancy-Driven Flow in a Square Enclosure with Variable Viscosity Effects," *Proceedings of ASME, Heat Transfer Division-HTD, IMECE*, Vol. 317-1, pp. 61-68, 1995.
- [13] Mahdi, H. S., Wedgworth, K. R., "Natural Convection in a Square Cavity with Temperature Dependent Viscosity" *Proceedings of ASME, Heat Transfer Division-HTD, IMECE*, Vol. 317-1, pp. 81-95, 1995.
- [14] de Vahl Davis, G., "Natural Convection of Air in a Square Cavity: A Bench Mark Numerical Solution," *International Journal for Numerical Methods in Fluids*, Vol. 3, pp. 249-264, 1983.
- [15] de Vahl Davis, G., Jones, J., P., "Natural Convection in a Square Cavity: A Comparison Exercise," *International Journal for Numerical Methods in Fluids*, Vol. 3, pp. 227-248, 1983.
- [16] Strada, M., Heinrich, J., C., "Heat Transfer Rates in Natural Convection at High Rayleigh Numbers in Rectangular Enclosures: A Numerical Study," *Numerical Heat Transfer*, Vol. 5, pp. 81-93, 1982.